

Problems with Using the Hencky Equivalent Strain in Simple Shear

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Onaka has recently (2010) discussed the application of the Hencky equivalent strain to the description of large strains in simple shear in the Journal of the Japan Institute of Metals. We find this analysis in error and introduce below what we consider as the appropriate approach to this problem. We also explain why the Hencky strain is unsuitable for converting the experimental results obtained in large strain shear and torsion to loading along other strain paths. [doi:10.2320/matertrans.M2011086]

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1. Introduction

The Hencky equivalent strain has recently been recommended as appropriate for the description of large deformation shear.¹⁾ The author of this article has also claimed that our analysis of 1982²⁾ did not take the necessary factors into account for dealing with severe plastic deformation; he has further maintained that it significantly overestimates the equivalent strain.³⁾ We find that the Ref. 1) analysis is in error and in what follows outline the problems associated with this treatment.

2. Geometric Parameters Associated with Simple Shear

We begin by defining the strains and angles associated with simple shear. These are identified in Fig. 1(a),⁴⁾ where γ and $d\gamma$ represent the shear strain and shear strain increment, respectively. Here a circle of unit radius is inscribed in an elemental square, which is deformed in simple shear. This square also refers to an element on any section that is perpendicular to the radius in a torsion specimen. In the case of torsion, the Z and θ axes correspond to the longitudinal and tangential directions of the sample, respectively.

During deformation, the square becomes a parallelogram, within which an ellipse is inscribed. The ellipse is then fitted with major and minor axes, which play important roles in what follows. The angle of inclination of the major axis with respect to the θ axis is defined as α . The relationship between γ and α is in turn given by:⁴⁾

$$\tan 2\alpha = 2/\gamma. \quad (1)$$

The Hencky principal strains ϵ_1^h and ϵ_2^h are defined in terms of the lengths of the semi-major and semi-minor axes and can be represented as being oriented along these axes. The tensile principal strain is thus inclined at the angle α with respect to the θ axis.

When an increment of shear strain $d\gamma$ is applied, the parallelogram is sheared, see Fig. 1(a), and a new ellipse, with a smaller inclination angle $\alpha + d\alpha$ can be inscribed. As a result, the inclination of the incremented Hencky principal tensile strain is now also reduced by $d\alpha < 0$. The relation

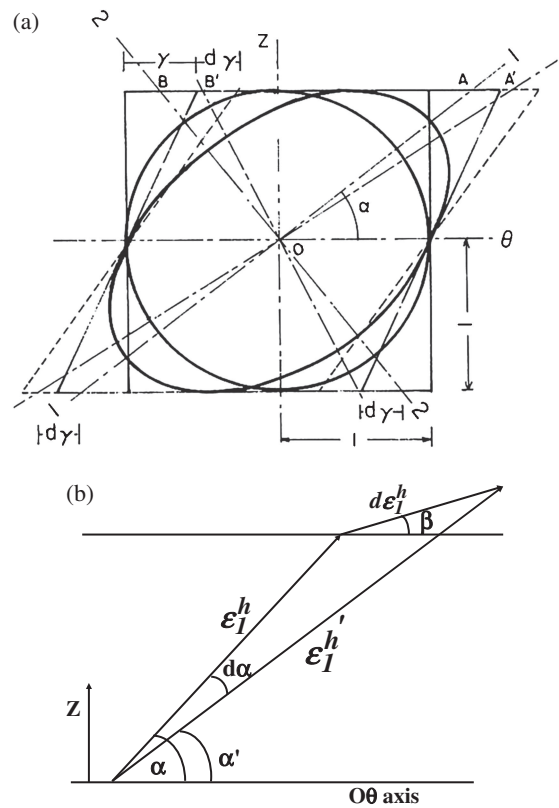


Fig. 1 (a) Deformation of an elemental square into a parallelogram during simple shear. The circle inscribed in the square deforms into an ellipse. α is the angle between the major axis of the ellipse and the circumferential direction $O\theta$.⁴⁾ (b) Definitions of the angles α and β , where ϵ_1^h and $d\epsilon_1^h$ represent the principal tensile components of the equivalent strain and equivalent strain increment, respectively. After an increment of strain $d\gamma$, the angle α decreases by an amount $d\alpha$ to α' and ϵ_1^h becomes $\epsilon_1^{h'}$. The angle of inclination of $d\epsilon_{eq}^h$ with respect to the (horizontal) $O\theta$ axis is β ; the positive directions for α and β are counter-clockwise here.

between ϵ_1^h and $(\epsilon_1^h + d\epsilon_1^h)$ is illustrated in Fig. 1(b), in which $d\epsilon_1^h$ and its angle of inclination β are also defined. It is of particular note that, because of the rotation of ϵ_1^h , $d\epsilon_1^h$ and ϵ_1^h are not colinear.

In the present context, the Hencky equivalent¹⁻⁶⁾ and principal strains can be defined as:

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$$\begin{aligned}\varepsilon_{eq}^h &= (1/\sqrt{3}) \ln\{[2 + \gamma^2 + \gamma\sqrt{(4 + \gamma^2)}]/2\} \\ \varepsilon_1^h &= (1/2) \ln\{[2 + \gamma^2 + \gamma\sqrt{(4 + \gamma^2)}]/2\} \\ \varepsilon_2^h &= -(1/2) \ln\{[2 + \gamma^2 + \gamma\sqrt{(4 + \gamma^2)}]/2\}\end{aligned}\quad (2)$$

and the Hencky equivalent and principal strain increments as:

$$\begin{aligned}d\varepsilon_{eq}^h &= (1/\sqrt{3}) \times (1/\sqrt{(1 + \gamma^2/4)})d\gamma \\ d\varepsilon_1^h &= (1/2) \times (1/\sqrt{(1 + \gamma^2/4)})d\gamma \\ d\varepsilon_2^h &= -(1/2) \times (1/\sqrt{(1 + \gamma^2/4)})d\gamma.\end{aligned}\quad (3)$$

The angle of inclination β of the principal tensile strain increment $d\varepsilon_1^h$, see Fig. 1(b), is in turn given by:

$$\beta = \tan^{-1}\{d(\varepsilon_1^h \sin \alpha)/d\gamma / d(\varepsilon_1^h \cos \alpha)/d\gamma\}.\quad (4)$$

The dependences of ε_{eq}^h and α on γ are illustrated in Fig. 2(a) and those of $d\varepsilon_{eq}^h/d\gamma$ and β on γ in Fig. 2(b). Of particular interest here is the difference $(\alpha - \beta)$ between the two angles of inclination; this is depicted in Fig. 2(c).

3. Application to Large Strain Torsion

We now apply this formalism to describe a simplified version of a torsion experiment carried out to a shear strain of $\gamma = 88$.⁵⁾ As an approximation, the measured data are plotted here in the form of a rigid-plastic shear stress vs. shear strain flow curve, see Fig. 3. In order to express these data in terms of their Hencky analogues, the shear strain must first be converted using eq. (2). Conversion of the stress, on the other hand, requires an assumption to be made regarding the invariance (or lack of invariance) of the incremental work. These matters are considered below.

3.1 Invariance of the plastic work and work increment

In plastic deformation, it is generally valid to state that the increment of mechanical work dW is an invariant. Thus, it is possible to write that:

$$dW = \tau d\gamma = \sigma_{eq} d\varepsilon_{eq}\quad (5)$$

where τ and $d\gamma$ are the shear stress and shear strain increment in torsion and σ_{eq} and $d\varepsilon_{eq}$ represent the von Mises equivalent stress and equivalent strain increment, respectively. When the stress and strain axes do not rotate, this conclusion can be extended to the Hencky formalism⁷⁾ to give:

$$dW = \sigma_h d\varepsilon_{eq}^h\quad (6)$$

where σ_h and $d\varepsilon_{eq}^h$ represent the Hencky equivalent stress and equivalent strain increment, respectively, and are assumed to be colinear. These expressions can also be written in integral form as follows:

$$\int \tau d\gamma = \int \sigma_{eq} d\varepsilon_{eq} = \int \sigma_h d\varepsilon_{eq}^h\quad (7)$$

We now make the tentative assumption, following Onaka,¹⁾ that the Hencky formalism can be employed even when the strain and strain increment axes are rotating, as they are in Figs. 1 and 2. In such a case, on the assumption that the work increment is invariant, we can write that:

$$\sigma_h = \tau d\gamma / d\varepsilon_{eq}^h\quad (8)$$

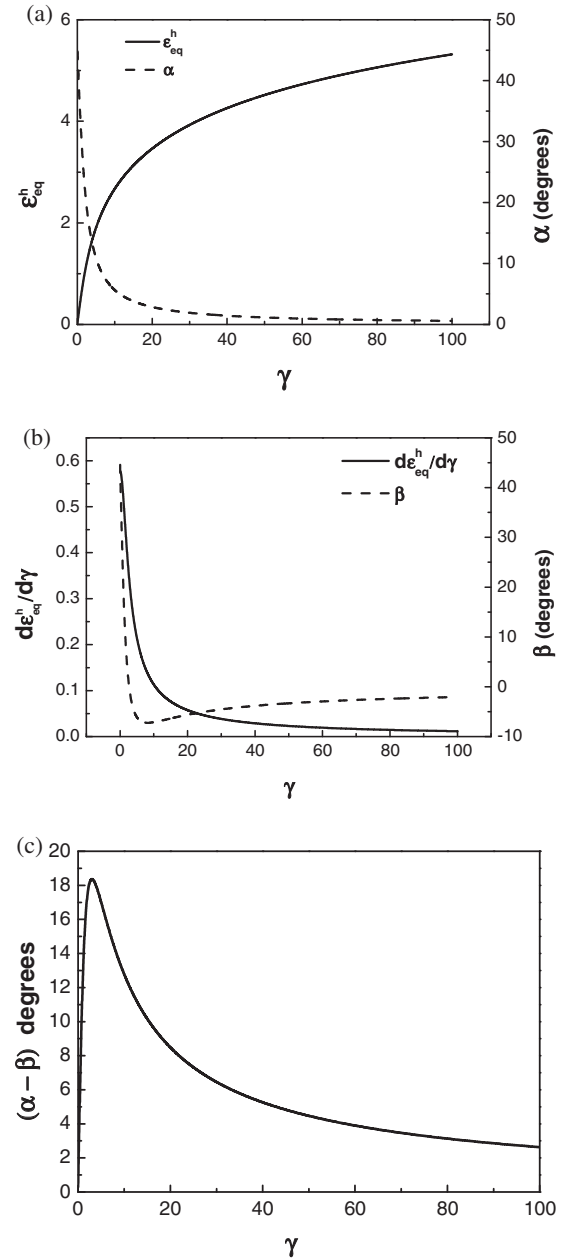


Fig. 2 (a) Dependence of the Hencky equivalent strain ε_{eq}^h on shear strain γ , eq. (2) (left hand axis) and of the angle α on γ , eq. (1) (right hand axis). (b) Dependence of the differential of the Hencky equivalent strain increment $d\varepsilon_{eq}^h/d\gamma$ on shear strain γ , eq. (3) (left hand axis) and of the angle β on γ , eq. (4) (right hand axis). (c) Dependence on γ of the difference in inclination of the principal tensile strain and strain increment axes $(\alpha - \beta)$.

where σ_h is assumed to be colinear with $d\varepsilon_{eq}^h$ (as when no rotation occurs). We use this expression to convert the schematic flow curve of Fig. 3 into its Hencky equivalent. The result of this operation is illustrated in Fig. 4.

Here it can be seen that the Hencky equivalent stress increases to 44 times its von Mises value, as given by the relation:

$$\sigma_h/\sigma_{eq} = \sqrt{(1 + \gamma^2/4)} = d\varepsilon_{eq}/d\varepsilon_{eq}^h = \dot{\varepsilon}_{eq}/\dot{\varepsilon}_{eq}^h.\quad (9)$$

Such stress values are clearly unrealistic and can be attributed to the invalidity of the assumption that the Hencky incremental work obeys the invariance relationships expressed above in eqs. (5) and (6).

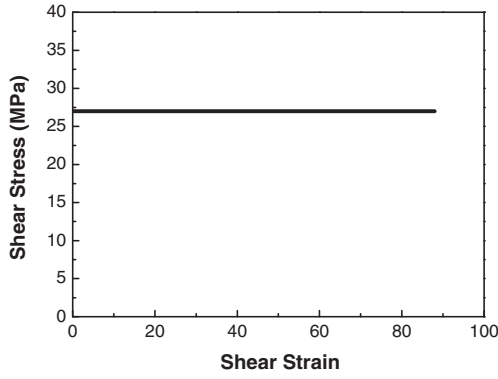


Fig. 3 Rigid-plastic approximation of the stress-strain curve of Ref. 5) determined in torsion on a 0.06%C plain carbon steel at 1200°C and a strain rate of 0.866 s⁻¹. The experiment was carried out to a shear strain of $\gamma = 88$.

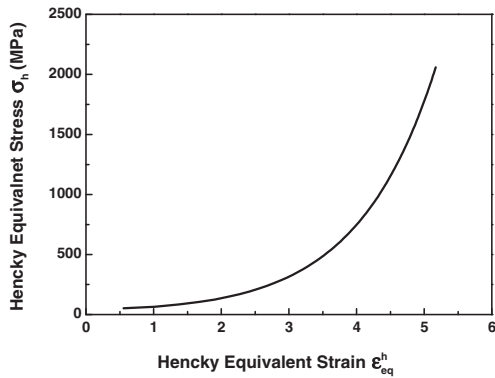


Fig. 4 The stress-strain curve of Fig. 3 converted into Hencky equivalent stress-Hencky equivalent strain on the assumption that the incremental work is invariant.

3.2 Constancy of the Hencky equivalent strain rate

A further problem arises when attempting to apply the Hencky formalism to torsion testing with regard to maintaining constant the equivalent strain rate.^{2,4)} This is a frequent requirement during mechanical testing. For the Hencky equivalent strain rate to remain constant during torsion, the shear strain rate would have to be increased during testing at a rate specified by the relation:

$$\frac{\dot{\gamma}}{\dot{\epsilon}_{eq}^h} = \sqrt{3} \sqrt{1 + \frac{\gamma^2}{4}}. \quad (10)$$

At shears of $\gamma = 50$ and 100, for example, the shear strain rate would have to increase by ratios of ~ 25 and ~ 50 over its starting value. This then entails a further problem associated with the rate sensitivity of materials at elevated temperatures. Given a rate sensitivity $m = 0.13$, for example, increasing the shear strain rate by factors of 25 and 50 would then produce unrepresentative flow stress increase ratios of 1.14 and 1.35.

3.3 The Hencky plastic work increment

Here, we take the rotations of the principal axes of the Hencky strain increments into account in order to calculate the incremental Hencky work without making any assumptions about work invariance. For this purpose, we define the Hencky work increment dW as:

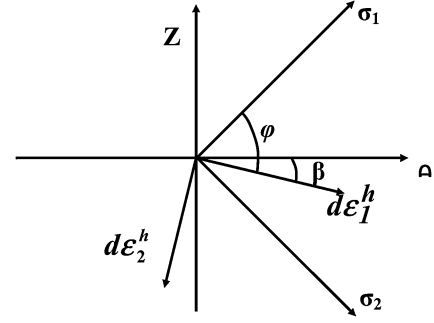


Fig. 5 Relation between the principal tensile and compressive stress axes (σ_1 and σ_2) and the principal tensile and compressive strain increment axes ($d\epsilon_1^h$ and $d\epsilon_2^h$). The angle between the two sets of axes is $\varphi = (45^\circ - \beta)$, where the angle β , measured *clockwise*, is negative in this case.

$$dW = \sigma_1 d\epsilon_{11} + \sigma_2 d\epsilon_{22} \quad (11)$$

where σ_1 and σ_2 are the principal stresses and $d\epsilon_{11}$ and $d\epsilon_{22}$ are the components of the Hencky principal strain increments resolved onto the stress axes. Thus, $d\epsilon_{11}$ and $d\epsilon_{22}$ consist of the following components, where $d\epsilon_1^h$ and $d\epsilon_2^h$ are the Hencky principal strain increments and φ is the angle between the two sets of axes (see Fig. 5):

$$\begin{aligned} d\epsilon_{11} &= d\epsilon_1^h \cos^2 \varphi + d\epsilon_2^h \sin^2 \varphi \\ d\epsilon_{22} &= d\epsilon_2^h \cos^2 \varphi + d\epsilon_1^h \sin^2 \varphi. \end{aligned} \quad (12)$$

The dependences on shear strain γ of the quantities $d\epsilon_{11}/d\gamma$ and $d\epsilon_{22}/d\gamma$ (see eq. (11)) are illustrated in Fig. 6(a). Here it can be seen that these terms change sign at a shear strain of 2.7.

Noting that $d\epsilon_1^h = -d\epsilon_2^h$, eq. (12) can be rewritten as:

$$\begin{aligned} d\epsilon_{11} &= d\epsilon_1^h \cos 2\varphi \\ d\epsilon_{22} &= d\epsilon_2^h \cos 2\varphi \end{aligned} \quad (13)$$

On substitution, the work increment can be defined as:

$$dW = \sigma_1 d\epsilon_1^h \cos 2\varphi + \sigma_2 d\epsilon_2^h \cos 2\varphi \quad (14)$$

and, bearing in mind that $d\epsilon_{eq}^h = 2/\sqrt{3} d\epsilon_1^h = -2/\sqrt{3} d\epsilon_2^h$ and that $\sigma_{eq} = \sqrt{3}\sigma_1 = -\sqrt{3}\sigma_2$, it can also be expressed as:

$$dW = \sigma_{eq} d\epsilon_{eq}^h \cos 2\varphi \quad (15)$$

Finally, given that $\varphi = 45^\circ - \beta$ (see Fig. 5), the work increment can be specified as:

$$dW = \sigma_{eq} d\epsilon_{eq}^h \sin 2\beta \quad (16)$$

Because of the changes of sign depicted in Fig. 6(a), dW changes sign (becomes negative) as well once a shear strain of 2.7 is exceeded, i.e. once the principal tensile strain increment dips below the horizontal (θ) axis, see Fig. 2(b). This is because the angle β in eq. (16), i.e. the inclination of the increment, has become negative.

Expressions (14) and (16) can now be integrated using the schematic stress-strain data of Fig. 3. The Hencky work calculated in this way is displayed in Fig. 6(b), where it is also displayed as a percentage of the total work specified by $\int \tau d\gamma$. It is readily evident that the work done to extend the major axis of the ellipse is only a few percent of the total work. That is, the Hencky formalism fails to take into

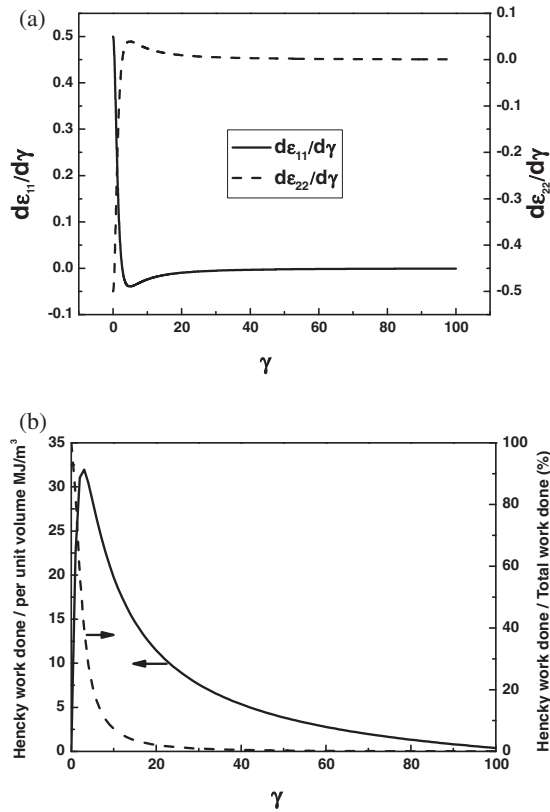


Fig. 6 (a) The dependences on shear strain γ of the quantities $d\epsilon_{11}/d\gamma = -d\epsilon_{22}/d\gamma$ (see eqs. (11)–(13)). Here it can be seen that these terms change sign at a shear strain of 2.7. (b) The integrated Hencky work done per unit volume (left hand axis) and the ratio (percentage) of the Hencky work over the experimental work $\int \tau d\gamma$ (right hand axis). Note that the Hencky work is only a small fraction of the total work and that it decreases with strain towards zero at $\gamma = \infty$.

account the largest part of the work done by the applied shear stress.²⁾ It is unable to do so because: (i) the principal stress and principal strain increment axes do not coincide; and (ii) the Hencky principal strain increments are only small fractions of either the von Mises or shear principal strain increments, both of which satisfy (with their conjugates) the requirements for the invariance of the incremental work. By contrast, the invariance of the work increment does not apply to the Hencky formalism when there is axis rotation because of factors (i) and (ii) above. This is the basic reason why the flow curve of Fig. 4 has such a strange (and invalid) shape. This physically unrealistic characteristic of the Hencky formalism is still another aspect of this approach that renders it unsuitable for describing large shear deformations.

3.4 Final note

A full tensor treatment of this problem, which takes the rotation of the principal strain axes into account on eq. (3), is provided in Ref. 6). This method leads to values for ϵ_{eq}^h , $d\epsilon_{eq}^h$, and β that differ slightly from those reported above. Nevertheless, this more rigorous approach leads to conclusions that are identical to those listed below.

4. Conclusions

(1) The Hencky principal strain and principal strain increment axes are not colinear and rotate at different rates. Neither set of axes coincides with that of the principal stresses.

(2) Because (i) the Hencky principal strain increment axes are not coaxial with the principal stress axes, and (ii) the Hencky principal strain increments are so much smaller than the principal shear strain increments, the incremental Hencky work only represents a small part of the total work. As a result, it cannot obey the rule regarding the invariance of the plastic work increment.

(3) When work increment invariance is nevertheless assumed to be valid, the resulting stress-strain curves are unrepresentative of the material behaviour.

(4) As stated by Nadai in 1937,⁸⁾ because of the continuous rotation of the Hencky principal strain axes (as well as the contrasting rotation of the Hencky principal strain increment axes), the Hencky formalism is not appropriate for use in describing the results of simple shear and torsion testing experiments.

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